

apply a marine■inspired anti■biofouling micro■textured surface (e.g., shark■skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin■binding thrombus

Executive Technology Brief

Domain: phage_microbio

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000009: Injury [SLIP-CATH BEACON TIP CATHETER]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'Polymeric Fibrous Materials for Procoagulant and Anticoagulant Applications: A Review of Molecular Blood-Material Mechanisms and Strategies.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: biofilm formation

Architecture: PROPOSED architecture (no physical system exists): phage loading / immobilization layer; release or contact-kill mechanism; surface substrate (coating architecture); stability preservation element — realizing: apply a marine■inspired anti■biofouling micro■textured surface (e.g., shark■skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin■binding thrombus

Governing Model: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

BIO-002: Exponential population dynamics. Expression: $N = N_0 * \exp(r * t)$. Applicability to this invention: CONDITIONAL; holds under: unlimited substrate; constant net rate over the window. Assumptions: kill kinetics on an immobilized surface are not well-mixed growth – re-measure before quantitative use. Engaged variables: NOT ESTABLISHED.

BIO-003: Surface-limited adsorption kinetics. Expression: $dN_{\text{infected}}/dt = k_{\text{ads}} * N_{\text{phage}} * N_{\text{bacteria}}$. Applicability to this invention: CONDITIONAL; holds under: well-mixed suspension; constant k_{ads} . Assumptions: surface-immobilized phage dosing is not well-mixed – use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific. Engaged variables: NOT ESTABLISHED.

BIO-001: Multiplicity of infection. Expression: $MOI = N_{\text{phage}} / N_{\text{bacteria}}$. Applicability to this invention: CONDITIONAL; holds under: well-mixed system; planktonic or defined-surface assay. Assumptions: surface-immobilized phage dosing is not a well-mixed MOI – use contact-area density instead (MODEL_DERIVED

adaptation). Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.